

Every receipt, sealed.



Merchant-signed receipts become shares of the brand you bought from, verified in Rust on Arbitrum Stylus.



Receipts stopped being evidence

 STOCKBACK

Any reward, refund or warranty flow that trusts a photo now pays out on forgeries.

50.1%

of people pick the AI-edited receipt in a side-by-side test. That is chance.

0.53–
0.60

AUC of the best forensic detectors on those forgeries. Barely better than guessing.

< 1 s

and a few cents to change one number on a receipt photo.

Wu et al., “When the Forger Is the Judge: GPT-Image-2 Cannot Recognize Its Own Faked Documents”, arXiv:2604.25213 (2026).

Points don't feel like ownership

 STOCKBACK

Loyalty programs are closed, confusing and can be changed under you.

6.7 /
14.8

programs people actually use, out of the ones they belong to.

73%

of consumers find loyalty programs too complicated.

~60%

of coalition loyalty programs fail within ten years.

Bond Brand Loyalty Report 2020, as cited in Oamen et al., "Orchestrating Rewards in the Era of Intelligence-Driven Commerce", arXiv:2512.00738.

Seal the receipt. Own the brand.



読

Scan

The merchant's till signs the receipt. You scan its QR. No photo needs to be trusted.

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Prove

The signature is checked, the receipt is counted once, and Stylus verifies the claim on-chain.

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Own

You receive shares of that brand's vault. No admin can dilute, freeze or expire them.

From receipt to ownership



Six steps. The last three happen inside one transaction on Robinhood Chain.

1



Till signs

The merchant signs every field: merchant, brand, receipt ID, amount, time, expiry.

🔑 Merchant key

2

You scan

The receipt travels as a QR code. Nothing personal is in it.

3

Attester checks

Verifies the merchant signature, expiry, brand and prior use. Then signs a claim of hashes only.

🔑 Attester key

4

Stylus verifies

Rust on Arbitrum checks the Ed25519 signature, at up to 9.2x less gas than Solidity.

5

Registry decides

Burns the receipt's nullifier, applies eligibility, caps and the sponsor budget.

6



You own

Shares of the brand's vault are minted to your wallet. No admin can touch them.

Off-chain, in seconds

One transaction: `submitClaim` on Robinhood Chain testnet

Not every proof proves the same thing

STOCKBACK labels every claim with the evidence behind it. A photo is never treated as proof of purchase.

1

[Live demo](#)

Merchant-signed receipt

The receipt is unaltered since the merchant signed it. Merchant simulated in the demo.



2

[Live demo](#)

Attested photo or OCR

The attester signed what it was given. A real purchase is not independently established.

3

[Roadmap](#)

Payment or order proof

Provenance from the payment or order source itself, for example zkTLS.

Why it matters: people pick an AI-edited receipt at chance (0.501) and forensic detectors reach AUC 0.53–0.60 (Wu et al., arXiv:2604.25213, 2026).

Who holds which key

Each party can only do its own job. No key ever reaches the browser.



If the attester key leaks

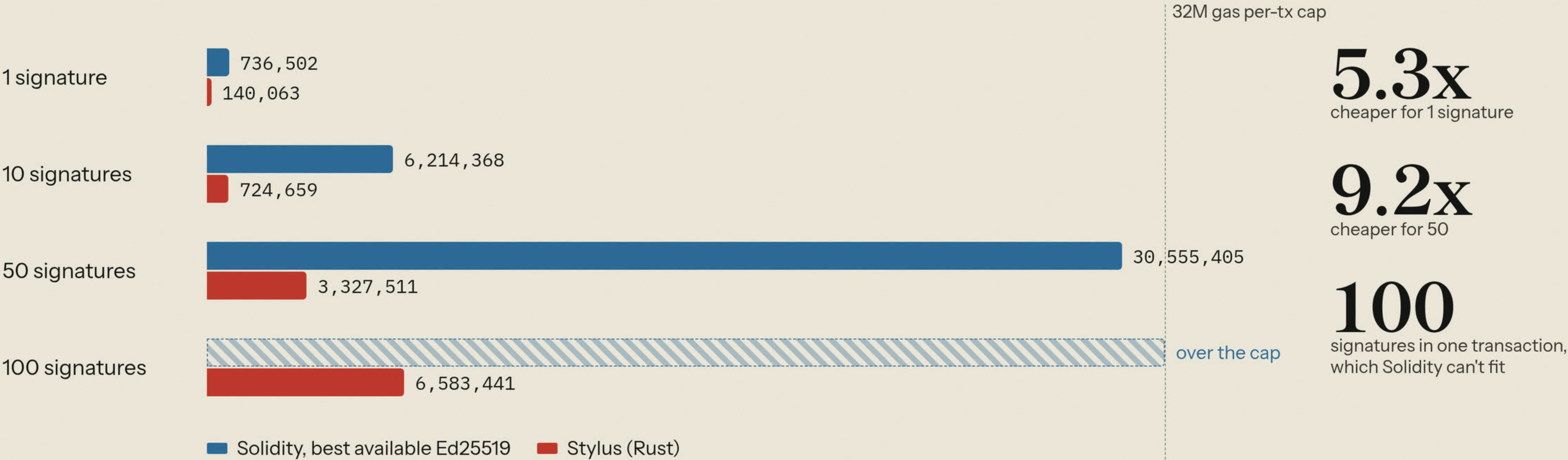
At most 100,000 demo units per brand per day, until the sponsor budget runs out. Existing holders are unaffected. The key can be rotated on the Stylus allowlist without a redeploy.

What this does not prove

That one person holds one wallet. Replay is blocked per receipt; Sybil resistance is future work. The QR is a bearer token, like a paper receipt.

9.2x less gas with Stylus

Strict Ed25519 verification of attestation signatures. Gas per transaction, measured on Robinhood Chain testnet.



eth_estimateGas, identical calldata, both verifiers deployed on chain 46630. ECDSA through the EVM precompile is cheaper still; Stylus earns its place for schemes the EVM lacks. Data: benchmarks/results/BENCHMARKS.md

Built, deployed, tested



Everything below runs today on Robinhood Chain testnet.

9

contracts: 8 Solidity plus a Rust
verifier on Stylus

83

Foundry tests, including fuzz and
invariants

47

more tests: 5 Rust, 20 receipt-
signing, 22 live integration

9.2x

less gas than Solidity for Ed25519

Registry

0x4273b12cD4A65c2180d4e65Bcb4254825cE64120

Stylus verifier

0x9ae8a390121ba71545e9923b333d60e7e3ccd3bd

Merchant-signed claim

0xabd3...5388

What's real, what's next



We'd rather show the edges than hide them.

Implemented

- Merchant-signed receipts (simulated merchant)
- Photo and OCR attestation
- One claim per receipt
- Caps and sponsor budgets
- Stylus Ed25519 verification
- Admin-less ERC-4626 vaults
- USDG funding adapter

Next

- Hold period with refund voiding
- Real merchant keys in a POS or HSM
- Merchant key registry and revocation
- k-of-n attestors, rotation runbook
- Stronger Sybil resistance
- Multisig and timelock

Not claimed

- Merchant partnerships
- Users or traction
- Measured fraud reduction
- Retention effects
- Real securities: all brand assets are testnet mocks



Scan. Prove. Own.



stockbacks.vercel.app

Live on Robinhood Chain testnet.
github.com/notwen123/STOCKBACK